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Abstract

Brief summary: This manuscript introduces Detector-Plane Imaging (DPI), a measurement-plane approach to imaging pre-measurement coherence and decoherence in imprint fields. We present simulation methods, experimental proposals, and preliminary results demonstrating emergent structures in nonlinear scalar-field evolution.

Introduction

Motivation and background: situate DPI with respect to camera-back physics, imprint-field hypothesis, and non-linear scalar-field dynamics. Outline contributions and structure of the paper.

Methods

1. Simulation framework: lattice discretization, ϕ^4 dynamics, time-stepping scheme (RK4/Leapfrog), boundary conditions. 2. Detector-plane projection: define detector plane geometry, sampling, temporal gating, and measurement operator. 3. Numerical diagnostics: energy density, coherence metrics, Fourier-space mode analysis, CHSH-type entanglement tests (if applicable). 4. Code and reproducibility: reference repository and environment (Python, NumPy, SciPy, JAX/PyTorch optional).

Results

Summarize key simulation outputs: imprint formation, lattice-pinned solitons, oscillons, coherence maps on detector plane, CHSH-style statistics. Include figure placeholders below.

Coherence

Coherence in DPI refers to sustained phase alignment and structured oscillatory order within the nonlinear scalar field before measurement intervenes. When the field's internal dynamics remain rhythmically coupled, extended regions evolve in step, generating smooth gradients, stable interference-like contours, and narrow mode distributions. These patterns reveal the field's hidden architecture: a quiet, collective choreography that precedes the discreteness of detection.

Methods Used to Analyze Coherence

Detector-Plane Sampling $\phi(x, y, t)$ and its energy density $T_{\mu\mu}$ are recorded at regular time steps on $z = z_{\text{det}}$. This allows coherence to be captured as it forms, not inferred afterward.

Two-Point Correlation Measures $C(r) = \langle \phi(x)\phi(x+r) \rangle$ Long correlation lengths indicate persistent coherence.

Phase-Gradient Tracking Spatial phase maps are computed; smooth gradients imply synchronized oscillations.

Fourier Mode Narrowing Coherent states show prominent low-k peaks and suppressed high-k scattering.

Temporal Stability Window Time-gated sampling isolates intervals where coherence stabilizes before nonlinear scattering disrupts it.

Outputs Observed for Coherence

- Structured Coherence Rings Bright, symmetric contours on the detector plane created by phase-aligned oscillations. These rings expand, contract, or breathe in rhythmic patterns.
- Long-Range Spatial Correlations Clear domains of ordered amplitude patterns spanning significant fractions of the grid.
- Narrow Spectral Peaks Fourier-space plots show concentrated energy in a small number of modes.
- Stable Oscillon or Soliton Signatures Regions of periodic self-contained oscillation that leave crisp, persistent imprints.

These outputs allow coherence to be identified visually, quantitatively, and temporally, forming the diagnostic backbone for pre-measurement order in the DPI approach.

Decoherence

Decoherence marks the erosion of the field's orderly phase structure as nonlinear couplings redistribute energy across many modes. Oscillatory domains become ragged, correlations shrink, and once-distinct spatial patterns dissolve into disorder. Decoherence is not an abrupt collapse but a continuous unraveling — a soft breaking of internal synchrony long before the detector enforces discreteness.

Methods Used to Analyze Decoherence

Phase Variance Growth Local phase variance maps track the loss of alignment across neighboring points.

Correlation-Length Decay $\xi(t)$ is computed from $C(r)$. Its contraction over time indicates decohering behavior.

Fourier Broadening Analysis Scattering into higher-k modes reveals increasing dynamical disorder. Temporal Slope Tracking Decoherence phases often show rapid changes in amplitude envelope and energy flow; dt-resolved sampling captures these transitions.

Event-Triggered Evaluation Collisions between oscillons, soliton decay, or turbulence on the lattice trigger targeted analysis windows to quantify decoherence onset.

Outputs Observed for Decoherence

- Fragmentation of Coherence Rings Smooth contours crack into uneven arcs and eventually dissolve into noise-rich patches.
- Patchy, Stochastic Detector-Plane Textures Once-ordered intensity maps become irregular and speckled.
- Spectral Spreading Fourier power spreads across many modes; dominant peaks flatten and blur.
- Rapid Collapse of Correlation Length $\xi(t)$ exhibits sharp downward inflections when nonlinear scattering overwhelms order.
- Chaotic Oscillon Dynamics Previously stable oscillons begin to wobble, merge asymmetrically, or disintegrate, producing high-frequency flicker on the detector plane. These outputs provide a clear dynamical signature of how the field loses its internal coherence, allowing DPI to chart the entire decay pathway rather than only the end-state distribution.

DOUBLE-SLIT INTERFERENCE (Using DPI Field)

Double-Slit Illumination Using the DPI Field

To test whether the detector-plane imaging (DPI) field carries physically meaningful spatial coherence, we use it as the illumination source for a classical double-slit aperture. The DPI amplitude snapshot is multiplied by a two-slit transmission mask, producing a spatially structured complex field incident on the slits. Unlike idealized plane-wave illumination, the DPI field contains non-uniform amplitude rings and spatial phase correlations inherited from the underlying nonlinear field dynamics.

Coherent Illumination and Interference Formation

When the DPI field retains coherence across the slit separation, the resulting far-field pattern shows high-contrast interference fringes. The coherent Fraunhofer output is displayed in Figure I2 (ds_dpi_intensity_coherent_best.png).

Key features include:

A bright, sharply defined central fringe.

Multiple alternating maxima and minima.

A double-band structure, formed by the convolution of:
the classical double-slit diffraction envelope, and
modal ring features from the DPI illumination.

This structured pattern verifies that the DPI field preserves phase relations across distant points on the detector plane.

Quantitatively, the cross-section taken along $k_x k_x$ (Figure I4, file ds_dpi_crosssection_best.png) reveals deep fringe minima approaching zero, yielding near-unity visibility. This confirms that the DPI field acts as a highly coherent illumination source.

Decoherence and Loss of Interference Contrast

To study how decoherence affects interference, controlled phase noise is added to the DPI illumination before it reaches the slits. This disrupts the mutual phase relationship between the two slit regions.

The resulting far-field pattern under partial decoherence is shown in Figure I3

(ds_dpi_intensity_partialcoh_best.png).

Compared to the coherent case, one observes:

Reduced fringe contrast.

Lifted minima in the central interference region.

Partial suppression of the double-band envelope.

These changes are captured clearly in the overlaid cross-section (Figure I4, ds_dpi_crosssection_best.png), where the decohered curve exhibits shallower minima and softened modulation.

This behavior matches classical coherence theory: as the mutual coherence function decays, the interference term weakens correspondingly.

Interpretation: What the Double-Slit Reveals About DPI

The double-slit experiment serves as a sensitive probe of the DPI field's coherence:

With coherent DPI illumination, classical interference appears with near-perfect visibility (Figure I2).

With decohered illumination, interference persists but the contrast is reduced (Figure I3).

The cross-section (Figure I4) provides a compact quantitative measure of coherence loss.

Together these results confirm:

The DPI field carries structured but robust spatial coherence, capable of driving classical interference phenomena and responding predictably to decoherence protocols.

The composite 4-panel summary (Figure I5) illustrates the entire pipeline: illumination → aperture → coherent interference → decohered interference → quantitative profile.

4. Photon–Oscillon Correspondence and the Origin of DPI

A key conceptual breakthrough enabling DPI was the recognition that the spatial structure of a quantum photon wavefront and the intensity profile of a nonlinear oscillon share the same geometric morphology. Both behave as localized, long-lived, coherent field packets:

They preserve internal phase order.

They propagate without significant dispersion.

They exhibit envelope–carrier structure.

They generate patterned spatial imprints when sampled at a plane.

This correspondence is not a claim of ontology — photons are not oscillons — but a geometric analogy that explains why the detector-plane in quantum experiments consistently shows structured coherence patterns long before the final “collapse.” Oscillons provide a classical stand-in for the coherent pre-measurement state of a photon.

Once this parallel was recognized, the interpretation of detector-plane data changed entirely.

Historically treated as a set of independent hits, the detector plane instead becomes a passive sampling surface that records the evolving imprint of a coherent field. DPI formalizes this reinterpretation.

5. Mathematical Framework of DPI

DPI requires only a minimal mathematical structure to make its measurement interpretation precise. Field Evolution We simulate a coherent excitation using a nonlinear scalar field model that supports oscillon solutions. The detector-plane intensity is simply the field squared:

Intensity = $\phi^2(x, y, t)$.

Accumulated Intensity To mimic single-photon interference buildup, we take the average of multiple frames:

Accumulated intensity = $(1/N) \times \text{sum of frames}$.

Coherence Length The field's degree of spatial order is quantified by how quickly the dominant Fourier component decays. The resulting scale is the coherence length $\xi(t)$, which decreases smoothly during decoherence.

Envelope Breathing The RMS radius of the intensity distribution gives the envelope width $\sigma(t)$, which oscillates due to nonlinear carrier–envelope coupling.

Double-Slit Propagation To test interference, we apply a double-slit mask to an imprint-field frame and propagate it to the far field via Fourier transform. High coherence → strong banded interference

Partial coherence → washed-out visibility

This small mathematical toolkit is enough to define DPI rigorously and reproducibly.

6. Results

6.1 Evolution of the Imprint Field on the Detector Plane

Figure Set 1 displays four snapshots of $\phi^2(x, y, t)$ at increasing times. They reveal a smooth evolution:

Early regime: clear concentric coherence rings

Peak coherence: maximal contrast and symmetry

Onset of decoherence: broken rings and local irregularities

Pseudo-3D imprint: showing the full envelope structure

These patterns match the qualitative signatures seen in bright coherent or partially coherent beams in nonlinear optics experiments.

Figure 1. Evolution of the imprint-field geometry on the detector plane. From left to right: (A) Early coherent rings with strong spatial order. (B) Peak coherence with maximal contrast. (C) Onset of decoherence, visible through ring distortion and added noise. (D) Pseudo-3D imprint proxy illustrating envelope structure.

Figure 1: DPI Imprint Evolution

6.2 Accumulated Intensity and the Single-Photon Buildup Analogy

Figure Set 2 shows cumulative averages over many frames. The pattern evolves from:

A diffuse central blob

Emerging faint fringes

A well-defined interference-like structure

This is essentially identical to single-photon interference buildup videos recorded by Aspden (2016) and Kolenderski (2014). DPI reframes this buildup as the gradual revelation of imprint-field structure rather than the accumulation of independent quantum “dots.”

Figure Set 2 : Panel Accumulated Intensity

Accumulated imprint-field intensity across increasing numbers of frames. From left to right: (1) ~10 frames: dominated by noise; fringe structure barely visible. (2) ~40 frames: rings begin to emerge. (3) ~100 frames: strong concentric structure appears. (4) ~180 frames: final high-contrast interference pattern. This buildup matches the gradual emergence of single-photon interference observed experimentally.

Figure 2: Panel Accumulated Intensity

6.3 Double-Slit Illumination and Interference Visibility

Figure Set 3 shows:

a coherent imprint-field frame

the same frame multiplied by a double-slit mask

the far-field interference pattern

the far-field pattern after partial decoherence

Strong coherence produces sharp banding; decoherence washes out visibility. This matches standard optical experiments where path information or noise reduces interference contrast.

Figure 3A — DPI illumination (coherent)

Early-stage oscillon imprint showing strong spatial coherence and concentric ring structure.

Figure 3B — Double-slit mask

Binary aperture selecting two narrow vertical regions of the coherent imprint field.

Figure 3C — Far-field (Coherent)

Structured high-visibility interference pattern produced by coherent illumination through the slits.

Figure 3D — Far-field (Decoherent)

Reduced-contrast diffraction pattern showing loss of fringe visibility under partial decoherence.

Figure 3: Double-Slit Illumination and Interference Visibility

6.4 Quantitative Diagnostics: $\xi(t)$, $\sigma(t)$, and $S(k)$

Figure Set 4 contains:

$\xi(t)$: coherence length decreasing over time

$\sigma(t)$: oscillating envelope width

$S(k)$: the structure factor at peak coherence

Cross-sections: showing loss of far-field visibility

These diagnostics demonstrate that DPI allows coherence to be directly measured at the detector plane — something standard interpretations treat as inaccessible.

Fig. 4A — Correlation length $\xi(t)$

$\xi(t)$ remains essentially constant, showing that global coherence is preserved during the early nonlinear evolution.

Fig. 4B — Oscillon breathing $\sigma(t)$

$\sigma(t)$ exhibits a clear breathing cycle, revealing periodic expansion and compression of the imprint-field core.

Fig. 4C — $S(k)$ at peak coherence

The Fourier-space intensity shows a structured, symmetric $S(k)$ pattern characteristic of a highly coherent nonlinear field.

Fig. 4D — Far-field cross-section

The far-field cross-section displays a deep central minimum and symmetric sidelobes, reflecting strong spatial coherence.

7. Interpretation

The DPI simulation reveals a consistent story: the detector plane is not merely a statistical endpoint but a surface that records a live, continuous projection of the coherent field.

Key implications:

Collapse is not instantaneous; it is preceded by measurable coherence decay.
Interference patterns are encoded in the imprint-field geometry long before the final detection event.
Decoherence manifests as a gradual restructuring of $\phi^2(x, y, t)$.

DPI turns the detector-plane into a coherence microscope.

All results in this paper arise solely from classical field dynamics. We make no claim of entanglement, nonlocality, or Bell inequality violations. Quantum DPI will be explored separately.

8. Experimental Realization of DPI with Existing Instruments

DPI is already achievable with standard laboratory hardware. The imprint-field structures simulated here are visible on:

1. Intensified sCMOS / ICCD cameras

Used in high-gain PDC and OPO experiments; they capture ring-like coherence modes and breathing structures identical to the imprint-field frames.

2. EMCCD cameras

These record single-photon buildup patterns matching DPI accumulated intensity.

3. SPAD arrays

Time-tagged photon detection maps directly onto DPI's cumulative imprint logic.

4. Consumer high-speed CMOS sensors

Even smartphones reproduce DPI far-field patterns when placed behind double slits with structured light.

Why this works

All these sensors measure intensity at a fixed plane without imaging optics, meaning they naturally implement DPI's geometry.

Immediate test protocol

Place a camera directly in a coherent beam.

Capture short-exposure frames.

Observe ring and breathing patterns.

Insert a double slit.

Measure visibility changes.

DPI requires no new equipment — only a reinterpretation.

Optional: Insert Figure Set 5 comparing real PDC data with DPI simulations.

9. Limitations and Scope

This paper presents DPI strictly in the classical-field regime. We avoid all claims regarding:
entanglement

nonlocal correlations

Bell inequality violations

photon quantization

These require a true quantum-field DPI model and two-detector-plane analysis, which will be developed in follow-up work.

10. Outlook: Toward Quantum DPI

A natural extension is to quantize the imprint field and apply DPI to multi-plane measurements. This would allow imaging of spatial entanglement, correlation decay, and potentially Bell-violating structures directly on detector planes.

This will be explored in a dedicated companion paper titled:

"Quantum DPI: Imaging Entanglement Through Detector-Plane Correlations."

11. Conclusion

Detector-Plane Imaging provides a new framework for understanding measurement. By treating the detector plane as an imaging surface rather than a statistics counter, DPI reveals the evolving spatial coherence of a coherent field prior to collapse. Oscillon-based simulations reproduce the same structures seen in optical experiments, demonstrating that existing laboratory cameras have already been recording imprint-fields — they were simply never interpreted as such.

DPI reframes measurement as a continuous dynamical process and establishes a foundation for future work in classical and quantum coherence imaging.

Discussion

Discussion

Detector-Plane Imaging (DPI) reframes the role of the detector in optical and field-theoretic experiments. Rather than treating the camera as a passive endpoint that registers discrete photon events, DPI shows that the detector plane itself contains a dynamically accessible record of field

coherence. The sequence of ring formation, breathing, fragmentation, and eventual decoherence observed in the imprint field is not a computational artifact: it is a direct diagnostic of internal synchrony within the evolving nonlinear scalar field.

The oscillon analogy demonstrates that coherence-driven spatial ordering emerges naturally in nonlinear dynamics without invoking quantum mechanical assumptions. Yet the same spatial signatures—structured interference, visibility decay, and cumulative fringe buildup—appear in laboratory experiments traditionally interpreted through single-photon language. Heralded-photon double-slit experiments (Aspden et al., Kolenderski et al.) show the same gradual progression from granular noise to high-visibility interference that DPI reproduces in purely classical field simulations. This alignment suggests that a significant portion of what we call “wave–particle duality” reflects the detector-plane manifestation of underlying coherence, regardless of whether the source is classical, nonlinear, or quantum.

Importantly, DPI provides a way to quantify decoherence as a continuous dynamical process through correlation-length decay $\xi(t)$, spectral narrowing $S(k)$, and aperture-filtered visibility transitions. These diagnostics exist at the level of the field—not the particle record—and therefore generalize across experimental platforms. Modern cameras (ICCD, EMCCD, SPAD, sCMOS) already record the raw spatiotemporal features DPI identifies; the framework presented here merely provides the missing interpretation.

DPI thus establishes a measurement-plane perspective in which coherence, not particle identity, dictates the morphology of observed interference patterns. This suggests a broader conceptual shift: imaging coherence directly at the detector surface may be as fundamental as counting individual detection events. The method points naturally toward a new class of coherence-aware instruments capable of resolving dynamical pre-measurement structure rather than only final detection outcomes.

Conclusion

Conclusion

This manuscript introduced Detector-Plane Imaging (DPI), a measurement-plane approach that maps the formation and decay of pre-measurement coherence in a nonlinear scalar field. Through oscillon-driven imprint dynamics, DPI demonstrated that emergent field structures can act as robust coherence sources capable of producing high-contrast interference patterns. The resulting behavior aligns closely with time-resolved single-photon interference experiments, indicating that many features typically attributed to wave–particle duality are already encoded in the coherence structure visible directly at the detector plane.

By interpreting decoherence as a continuous dynamical process—rather than an abrupt measurement-induced collapse—DPI positions the detector as an active coherence imager. This conceptual shift provides a foundation for a more general framework of Quantum Microscopy, in which spatial coherence is treated as a measurable physical field. The results presented here motivate the development of new sensor architectures designed to capture pre-measurement coherence, opening pathways toward enhanced imaging, improved diagnostics, and a deeper understanding of dynamical field behavior in both classical and quantum systems.